

Problem 1. Suppose that $F, G \in K[x]$, show that $\gcd(F, G) \notin K[x]^\times$ if and only if $\text{Res}(F, G) = 0$

Proof. Sufficiency: According to $\gcd(F, G) \notin K[X]^\times$, we suppose that $H = \gcd(F, G)$, then we have $H \mid F$ and $H \mid G$. So we can represent $F = AH$ and $G = BH$. For A, B , $\deg(A) < \deg(F)$, $\deg(B) < \deg(G)$. We can take $A, -B \in K[X]$ so that

$$BF - AG = ABH - ABH = 0 = \text{res}(F, G)$$

Necessity: According to $\text{res}(F, G) = 0$, so we know that $\exists A, B \in K[X]$ so that $\deg(A) < \deg(G)$, $\deg(B) < \deg(F)$, $A, B \neq 0$ and $AF + BG = \text{res}(F, G) = 0$. Then we have $AF = -BG$. We suppose that $\gcd(F, G) \in K[X]^\times$, with $AF = -BG$ we know that $F \mid B$. If $B \neq 0$, $\deg(B) \geq \deg(F)$, contradiction, $\gcd(F, G) \notin K[x]^\times$. If $B = 0$, then $A = 0$, contradiction, so $\gcd(F, G) \notin K[x]^\times$ $\square$